



The long-term effects of job training on labor market and skills outcomes in Chile

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ABSTRACT

Job training programs can be an effective policy for improving productivity and labor market outcomes in low- and middle-income countries. We report medium- and long-term impacts of a job training program for vulnerable workers in Chile on labor market and skill outcomes using experimental and administrative data. We estimate intention-to-treat effects, accounting for potential non-compliance by training providers in the assignment process, as well as local average treatment effects due to noncompliance with the assigned treatment status. Overall, the program shows limited effectiveness. While we find some positive impacts on labor income, the program fails to improve most labor outcomes. A detailed investigation of heterogeneous effects by course-type, training provider and course quality, and gender reveals that the positive income effects are more likely to be realized for males and participants in courses that were offered by high-quality providers. Evidence on mechanisms indicates that the program's limited impact on labor outcomes could, at least partly, be due to its limited effect on workers' skills.

1. Introduction

Although education coverage in Latin America and the Caribbean (LAC) has substantially expanded in the last thirty years, the region still faces the challenge of improving both the quality and relevance of its education. These deficiencies are reflected in the poor performance in standardized international tests, skills gaps reported by employers, high levels of informality, unemployment, and low productivity of countries in the region. In this context, job training has received a lot of attention in LAC as an effective policy aiming at redressing the lack of skills obtained during formal education and improving adults' labor outcomes (Escudero et al., 2019). Evaluations of such training programs show, in general, positive effects on different labor outcomes in the short-term (Attanasio et al., 2011; Card et al., 2011; Ibarrarán and Rosas Shady,

2009) that, according to the few long-term evaluations, do not improve over time (Attanasio et al., 2017; Kugler et al., 2015; Alzúa et al., 2016; Ibarrarán et al., 2018; Novella et al., 2024). In this paper, we evaluate the medium- and long-term effectiveness of a large training program in Chile on skills development and labor outcomes.

Chile is a pioneering country in LAC when it comes to implementing job training programs. However, there is scarce evidence about how effective these programs are in improving labor market outcomes. Available evidence from non-experimental evaluations show positive results on earnings and employment probabilities (Aedo and Pizarro, 2004; Centro Microdatos, 2006, Centro Microdatos, 2008) and larger effects for vulnerable groups, such as youths (Aedo and Pizarro, 2004) and less-educated workers (Novella et al., 2017). A randomized experiment gives us the opportunity to present, for the first time, evidence on

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the effectiveness of vocational training. The *Formacion para el Trabajo* (FOTRAB) program, which started in 2012 and persisted until 2018, primarily provided four to six months of classroom and on-the-job or technical assistance training to women and men between the ages of 18 and 65 living in vulnerable conditions.

This study contributes to the literature evaluating the long-term effects of training programs in developing countries. We provide the first experimental evidence on a large-scale training program aimed at individuals in vulnerable circumstances in Chile, without restricting eligibility by age-group or employment status. In particular, our study has three key features that make it a good complement to existing papers. First, by linking participants to employment records from administrative data, we are able to evaluate the long-term effects of training, four years after application. Second, by leveraging data from the training providers, we are able to explore whether the labor market effects vary based on the quality of the providers—a dimension that has been largely overlooked in previous studies.¹ Third, by using rich data collected on both cognitive and non-cognitive skills, we are able to examine a key mechanism through which training programs are expected to influence labor market outcomes: the accumulation of human capital.

This study is based on a sample of more than 5000 applicants to the FOTRAB program. These individuals were randomly assigned to either the treatment group, receiving the program, or the control group, in one cohort of applicants in 2013, using an over-subscription design. In the evaluation, we focus on 123 courses that were offered in three regions in Chile. The applicants were surveyed at three points in time. The first survey took place before training started, while the other two surveys took place in the post-training period 18 and 35 months after application. We further enrich our analysis by using administrative data, which serves as a robustness check for the validity of the self-reported labor market outcomes of survey participants.

We estimate intention-to-treat effects, carefully considering potential non-compliance by training providers in the treatment assignment process. Additionally, given noncompliance with the allocated treatment status, we also estimate local average treatment effects. We find that being assigned to, and participating in, the program has no effect on the probability of engaging in paid employment or being self-employed. Labor income increases in the short run, but such improvements disappear over time. When we distinguish the overall effect into the three different course types, we find that the most basic classroom training has a positive long-term effect on labor income. The course that combines classroom training with technical assistance shows positive, but not statistically significant, point estimates on being self-employed in the short-term and being formally employed in the long run. Courses that combine classroom training and on-the-job training are not effective at all. Analyzing heterogeneity, we find that the positive effects of the program on labor income are significantly larger for males, older individuals, and participants with lower estimated formal labor prospects. We also find that training from a high-quality provider has a larger positive effect on labor income.

We discuss various mechanisms that might help explain the largely null average effects of FOTRAB on labor market outcomes. First, we investigate the impact of the program on the skills development of participants. We find that while FOTRAB has a small positive impact on cognitive skills in the short run, it has no persistent effects on cognitive skills, nor any of the non-cognitive skill measures. Therefore, although the program aimed at improving workers' skills, it seems to have been largely ineffective in developing human capital. Second, we emphasize that the program followed a "supply-driven" approach, in which there was no consultation with industry partners to ensure that skills taught in

the program were relevant to the evolving demands of the local workforce.

This article is organized as follows: Section 2 describes the program and the design of the experimental evaluation. Section 3 presents the data used and analyses sample attrition. Section 4 analyses sample balance and non-compliance. Section 5 discusses the empirical strategy. Section 6 presents the results on labor outcomes, and Section 7 discusses heterogeneous effects. Section 8 examines potential mechanisms behind our results, while Section 9 concludes.

2. Program description and experimental design

Over the past three decades, Chile has emerged as one of LAC's fastest-growing economies, successfully reducing its poverty rate from 36 % in 2000 to 8.6 % in 2017. However, the country still faces important challenges in term of reducing high inequality and improving human capital. According to the World Banks' Human Capital Index, the country performs better than the LAC average but worse than the average of high-income countries in helping new generations achieve their human capital potential.² Compared to other OECD countries, Chile exhibits lower rates of secondary education completion, and lower scores in reading, literacy, math, and sciences.³ In 2012, the employment rate was 55.7 %, with much lower rates for females (43.4 %) and youth (31.1 %). Although unemployment was relatively low in the country (6.7 % in 2012), informal employment has persisted at around 35 % over the last decade.⁴ In this setting, training workers represents a policy alternative to improve employability in quality jobs.

2.1. Formación para el Trabajo (FOTRAB)

The origins of FOTRAB date back to 1997, when it was implemented as the *Programa Especial de Jóvenes* (PEJ), a subsidized training program aimed at improving job skills, labor market integration, and employability of young people (18 to 29 years) living in vulnerable conditions. Positive non-experimental evaluations of PEJ (Jara, 2001; Centro Microdatos, 2006, Centro Microdatos, 2008; Ministerio de Trabajo y Previsión Social, 2011) prompted expansion of the program.⁵ In 2011, it was extended to unemployed women aged 25–49 and both genders over 50, operating for a year under *Formación en Oficios* before being renamed *Formación para el Trabajo* (FOTRAB) in 2012. FOTRAB targeted individuals aged 18 to 65 in vulnerable conditions, identified as belonging to the first two quintiles of the *Ficha de Protección Social* (FPS).⁶ The program offered job training to all individuals living in vulnerable conditions independent of their employment status and young parents between 16 and 17 years old, but excluded tertiary education graduates, current students, and recent program participants.

² The World Bank <https://data.worldbank.org/country/chile> (accessed 17 December 2019).

³ OECD <https://data.oecd.org> (accessed 17 December 2019).

⁴ Inter-American Development Bank <https://www.iadb.org/en/sector/social-investment/sims/home> (accessed 17 December 2019).

⁵ Using non-experimental methods (before-after comparison, matching, and difference-in-difference), these evaluations find that PEJ is associated with a reduction in the probability of being unemployed and with an increase in the probability of being employed, being formally employed, and labor income.

⁶ Eligible individuals must have an FPS score of 11,734 points or lower. The FPS was, until 2016, the country-wide tool for targeting public social program. The exact calculation of the FPS score has not been made public, but the score is determined by assessing the income-generating capacity of working-age household members, considering factors like education, work experience, and regional economic conditions. Income from pensions and subsidies is also factored in, and household needs are calculated based on demographic characteristics. The score does not include ownership of assets but does also consider household size and any disabilities among members (Herrera et al., 2010).

¹ The paper by Galdo and Chong (2012) is the only other paper that we know of which analyzes the differential effects of training programs according to provider quality.

FOTRAB was financed by the Chilean government and administered by the National Training and Employment Service (SENCE). SENCE designed the curriculum and opened a call to private training providers, *Organismos Técnicos de Capacitación* (OTEC), that could apply for the implementation of the courses. In the call, SENCE specified the subject of training and the topics that must be included. The curriculum was built to improve participants’ human capital by providing them with cognitive and non-cognitive skills, as well as technical knowledge specific to their occupation. Through this, the program was expected to enhance employment prospects for unemployed individuals (including through self-employment), support the transition from informal to formal employment, and increase wages for those currently employed. OTECs were selected based on proposals for implementation, evaluated by SENCE, with the top performers hired and remunerated per participant. The first cohort entered FOTRAB in 2012 and the program was active until 2018.

Outreach activities for FOTRAB were primarily overseen by the regional offices of SENCE, with additional support from the communications office at SENCE headquarters. The OTECs also played a critical role, as they were incentivized to recruit applicants for the program. Although the specifics of promotion varied across regions, common channels included radio and TV advertisements, distribution of leaflets, and engagement through social media platforms like Facebook.

Eligible individuals could apply to a subsidized job training course from a set of available courses in each region. To do so, individuals visited registered OTECs, where they received detailed information about the courses offered, including their content and type (discussed below). With this information, eligible individuals were given the freedom to apply for any training course that they felt best suited their needs.

The average duration of the offered courses ranged between four to six months. The courses can be classified in three types (see Table 1). The first type is classroom training (CT), which took place in the classroom at the training provider and consisted of a minimum of 250 h of training. On average, classroom training courses lasted 500 h and individuals spent four hours per day in the classroom. At least 60 % of the total duration was devoted to teaching technical skills specific to the trained occupation, while the remaining time was spent teaching transferable skills (e.g., noncognitive skills, digital, and financial knowledge).⁷ Participants were provided an accident insurance, a daily subsidy of US\$6 for transport and food,⁸ and certificates or licenses in case the occupation required them. These benefits were also offered in the two other course types.

The second course type consisted of classroom training plus practical

job training (CJT). In addition to the classroom component, these courses offered 360 h of on-the-job training and job-finding support. For the practical training, each OTEC took on the responsibility of sourcing and reaching out to local businesses willing to host CJT participants. Following the classroom phase, all practical training took place at the premises of the respective firms. The focus of both CT and CJT courses was on securing participants paid employment.

The third course type consisted of classroom training plus technical assistance (CTA). In contrast to the other two types, these courses were designed for individuals who were interested in being self-employed.⁹ In addition to the content taught in the classroom, the curriculum covered at least 40 h of technical assistance (covering, for instance, topics like self-marketing, branding, and accounting) to improve self-employability and a subsidy of up to US\$404 for buying tools for those who finalized the technical assistance phase.

Although FOTRAB was the largest public training program for vulnerable individuals during its period of operation, there were other SENCE training programs operating at the same time. Among these, notable examples include the Training in Trades (CAP) program, which targeted vulnerable men and women aged 16 to 65 years old, aiming to increase their employability through comprehensive training in trades and subsequent job placement support. Additionally, on-the-job training programs (apprenticeships) focused on providing practical training components alongside theoretical education to enhance employability, particularly targeting unemployed individuals between 16 –24 years of age. Furthermore, the Training Bonus for Micro and Small Entrepreneurs offered training opportunities aimed at improving the skills and capacities micro and small business owners.

2.2. Experimental design

The experimental evaluation of the program was designed in 2012, under the framework of a loan between SENCE and the Inter-American Development Bank (IDB). The evaluation was motivated by the evolution and importance of the program, as well as the lack of experimental evaluations of training programs in the country. The experiment took place in one of the three cohorts of applicants to FOTRAB in 2013 (*Llamado* 22). The experimental design involved several steps. First, a sample of 123 courses offered in three regions (Biobío, Metropolitana, and Valparaíso) and organized by 12 training providers were selected. SENCE decided to implement the evaluation in the three regions with largest demand. The selection of courses was guided by two criteria: selected courses needed to represent the diversity of courses offered by FOTRAB and had to show enough demand in previous *Llamados* to ensure that randomization by excess of applications could be implemented. The selected courses represent 46 % of courses in the three regions (and 31 % of the total courses implemented in the country in that *Llamado*). Selecting courses from the three regions with the highest demand, and ensuring they represent the program’s diverse offerings, enhances the external validity of the evaluation to other cohorts of the FOTRAB program. The type of offered courses varied by region and provider. A large majority (67 %) of courses were CTJ courses, 20 % were CTA courses, and only 13 % were pure CT courses. The distribution of course types by region and OTEC is presented in Table A.1 in Appendix A.

As the second step, training providers were asked to prepare a list of eligible applicants equivalent to twice the capacity of each training course. Third, SENCE/IDB received the list of eligible candidates and randomly assigned them to the treatment or the control group.

Table 1
Training content and features by course type.

I. Classroom training (CT)	
(1) Classroom training (min. 250 h)	
(2) Accident insurance	
(3) Food and transportation subsidy (US\$ 6 daily)	
(4) Certificates and licenses	
II. CT plus job training (CJT)	III. CT plus technical assistance (CTA)
(5) On-the-job training (360 h)	(5) Technical assistance (min. 40 h)
(6) Job-finding support	(6) Subsidy for tools (US\$ 404)

Note: There are three different course types. Contents (1)-(4) are common to all three course types. CJT and CTA offers additional two components, respectively.

⁷ Training providers were allowed to vary the distribution of hours between these modules and teaching methodologies conditional on offering training on specific technical and transferable skills.

⁸ The 2013 average exchange rate of US\$1 = 495 Chilean Pesos (CLP) is used in the paper.

⁹ There was no formal process from the OTECs to identify applicants interested in self-employment. Instead, during the application process, individuals were informed that CTA courses catered specifically to those interested in self-employment. They were then given the autonomy to choose these courses if they felt they best suited them.

Randomization was realized within each course and stratified by gender. Overall, 5193 applicants were randomly assigned to the control and treatment group (columns 1 and 2 of Table 2). Fourth, the list of individuals assigned to the treatment group was sent to the training providers to proceed with their registration and to start the course. From applicants who were originally assigned to participate in the program, about 39 % never started or dropped-out early. Every time an applicant originally assigned to the treatment group was no longer available or dropped-out within the first 20 % of the training, the provider should have contacted SENCE/IDB who would randomly select a replacement out of the list of controls.

The implementation of the evaluation followed the experimental design, with the exception of the fourth step. When the OTECs received the lists of applicants assigned to the treatment groups, they were able to infer who was not assigned to treatment. This allowed them to skip the step where they had to contact SENCE/IDB to receive a randomized replacement and, contrarily to the design, directly contacted applicants originally assigned to the control group. Unfortunately, we do not know how often this happened and how the OTECs decided which individuals they offered a slot for being reassigned. As such, we carefully investigate whether original and realized assignment is random based on the applicants' information in our data (see Section 4). The advantage of our study is that we observe the baseline characteristics and outcomes for nearly 90 % of applicants. This allows us to carefully investigate potential selection in the re-assignment procedure.

We focus on the impact of being offered to participate in FOTRAB either during the original randomization or through re-assignment. The number of applicants assigned or re-assigned to the treatment and the control group are shown in columns (3)-(4) of Table 2. Due to the re-assignment procedure, the final sample size of the treatment group is larger than the control group. We define an assigned person as a participant if she or he passes the initial 20 % of the course. The numbers of participants, no-shows (the ones who never started or dropped out earlier) and non-participants are reported in columns (6)-(8). The compliance rate, defined as the share of participants among those assigned to a course, is 67 %.

3. Data collection and sample attrition

Applicants' information was collected at four points in time in face-to-face interviews: (i) basic data was collected at the time of application; (ii) a baseline questionnaire in 2013; (iii) a medium-term follow-up survey about 1.5 years after applying to FOTRAB; and (iv) a long-term follow-up survey in 2016, roughly 3 years after applying to FOTRAB. In addition, we use data from administrative records that track individuals four years after application to the program. This allows us to estimate short-, medium- and long-term impacts of the FOTRAB program.

3.1. Basic data and baseline survey

Basic demographic characteristics and information for confirming eligibility to FOTRAB were collected among all individuals who applied to the program. The baseline survey was carried out at the homes of each participant shortly before the training started. The survey was administered by the Observatorio Social at the Alberto Hurtado University and included information such as socio-demographic variables, information about household characteristics, education, skills, and self-reported labor market status.

A large advantage of the evaluation of the FOTRAB program is the extensive measurement of both cognitive and non-cognitive skills. Cognitive skills at baseline were measured using tests of literacy,

numeracy, spatial ability, and fluid intelligence. Three dimensions of non-cognitive skills were also measured at baseline: the Big-Five personality traits (openness to experiences, conscientiousness, extraversion, agreeableness, and neuroticism), locus of control, and self-esteem.¹⁰

3.2. Follow-up surveys

The first follow-up survey occurred between the end of 2014 and the beginning of 2015, on average 18 months after individuals applied to FOTRAB. The survey was conducted at the homes of each participant and was administered by the Faculty of Economics and Business at the University of Chile. The subsequent follow-up took place in 2016, on average 35 months after applying to the program. Data collection occurred at participants' homes and was administered by the Centro de Microdatos within the Department of Economics at the University of Chile. Similar to the baseline survey, information on personal and household characteristics, as well as self-reported labor market status, were collected in both points in time. The labor market information included the employment or self-employment status, contributions to the social security systems, monthly labor income, and the number of hours worked per month.

Again, cognitive and non-cognitive skills were measured in both follow-up surveys. For cognitive skills, similar tests as in the baseline survey were included in the first follow-up. These tests were not repeated in the second follow-up but were replaced by a Cognitive Reflection Test (CRT) (Frederick, 2005). For non-cognitive skills, the self-esteem test was repeated in the first follow-up and complemented by a GRIT measure (Duckworth and Quinn, 2009) and a Personal Competencies Scale (CPS) test (Brea, 2010). Similar to the cognitive skills, the second follow-up included only one test for non-cognitive skills – a measure of self-efficacy (Jerusalem and Schwarzer, 1992).

3.3. Attrition at baseline and follow-up surveys

The overall attrition rates are relatively low in this evaluation. First, we observe some attrition between application and the baseline survey, as 89 % of applicants were interviewed at the baseline. We perform regressions to investigate if the attrition rates significantly differ between applicants who were originally assigned to participate in the course and those assigned to the control group (see Table B.2 in the Appendix). We regress an indicator of participation in the baseline survey on an indicator for original assignment to the treatment group, course-by-gender fixed effects, and basic characteristics collected at the time of application. We find that the coefficients are close to zero and not significant, which suggests that attrition between application and the baseline survey is random.

Next, we focus on attrition in the follow-up surveys (columns 3–6 of Table B.2). Because reassignment happened after the baseline survey (but before the follow-ups), we regress an indicator of participation in the first or second follow-up survey, respectively, on an indicator for being either originally or re-assigned to the treatment group, course-by-gender fixed effects, and characteristics provided at application and in the baseline survey. We find a positive coefficient of assignment to treatment on the likelihood to participate in the first follow-up survey of 2–3 percentage points (pp) depending on whether we control for basic characteristics. Although the effects are not large, they could lead to a bias in the empirical results. We find no relation between treatment assignment and the likelihood to participate in the second follow-up survey. Analyzing the characteristics provided at application and in the baseline survey, we find that participation in the first follow-up survey is negatively correlated with formal employment and formal

¹⁰ Appendix B contains additional details about the information collected and the measures used to assess both cognitive and non-cognitive skills.

Table 2

Original randomization, realized assignment, and participation.

	Original Randomization		Realized Assignment			Realized Participation		
	Treatment Z* = 1	Control Z* = 0	Assigned Z = 1	Re-assigned	Not assigned Z = 0	Participant D = 1	No-show D = 0	Control
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
All	2631	2562	2631	1020	1542	2464	1187	1542
CT	330	321	330	106	215	311	125	215
CJT	1701	1675	1701	760	915	1561	900	915
CTA	600	566	600	154	412	592	162	412

Note: Z* indicates original randomization. Training providers had the opportunity to fill empty slots of assigned applicants that did not show-up or dropped-out early. Z indicates if an applicant was randomly assigned to participate in a course either by original assignment or by re-assignment. D indicates participation in a course. The evaluation is based on the realized assignment. CT is the abbreviation for classroom training, CJT is classroom plus job training, and CTA is classroom plus technical assistance.

salary in 2012. However, no such selection dynamics are found in the second follow-up survey.

4. Experimental balance and non-compliance

4.1. Balancing of applicants' characteristics: original treatment assignment

Randomization of participants into treatment and control groups was performed before the baseline survey, using application data. If the randomization of applicants was successfully implemented, the characteristics of those who are assigned to the control group and those who are offered to participate in training would be not statistically different. In Table C.1 in the Appendix, we present descriptive statistics and balance tests for both the full sample, as well separately for each type of course. Since randomization took place within training provider and course, and was stratified by gender, we control for course-by-gender fixed effects in all regressions. Furthermore, given the attrition discussed above, we present all sample diagnostic for individuals who have information on paid employment in both follow-up surveys (our analysis sample).

We observe that applicants were 32 years old on average and had an average poverty score of 5807. This indicates that, on average, they belonged to the poorest quintile of the population. 34 % of applicants were male, and 57 % were formally employed at some point between 2011 and 2012; however, only 46 % were formally employed in the year before the application. On average, participants were employed for 3.6 months in 2012 and had a monthly labor income of 124,000 Chilean Pesos. 42 % of the applicants are married or live in a partnership, and the average household size is four persons with two children living the household.

Applicants to the program accomplished, on average, 11.5 years of education - which is just below the 12 years needed to complete secondary education. The test scores for cognitive skills support the view of a middle- to high-skilled sample, as the achievement on the cognitive skill tests for literacy is close to 90 %, and the score on the tests for numeracy and spatial orientation is 75 %. Fluid intelligence is tested on a scale from 0 to 100 with an average score of 54. On average, applicants show a high internal locus of control and a high self-esteem score (34 points out of 40).

When we focus on the differences between those originally assigned to the treatment and control group, we find that the baseline characteristics are well-balanced. Differences are largely small and statistically insignificant. While those assigned to FOTRAB are less likely to be the head of the household, have slightly more years of education, and have a lower average conscientiousness score, a test for the joint significance of all baseline characteristics to predict treatment assignment is not statistically significant (p-value = 0.465). Therefore, despite the few significant differences, we conclude that the original assignment procedure was successfully randomly implemented in the full sample.

We next move on to analyzing the randomization for each course type separately (columns 4–9 in Table C.1). For each type, we observe mostly small and statistically insignificant differences between those assigned to treatment and control groups. Indeed, for all three course types, it cannot be rejected that the mean differences between treatment and controls in the baseline variables are jointly zero. Therefore, we conclude that the randomization was successfully implemented within each course type.

As previously mentioned, before randomization was implemented, individuals eligible for participation were given the freedom to select the training course that they felt best suited their needs. This process of self-selection led to notable differences in participant characteristics among the three course types. As expected, CTA courses attracted a higher percentage of self-employed individuals, with 38 % of applicants in the control group being self-employed at the outset, compared to only 16 % and 18 % for CT and CJT courses, respectively. Furthermore, applicants for CTA courses tended to come from relatively less wealthy households, as indicated by their average poverty score of 4864, compared to 6906 for CT participants and 6012 for CJT participants. In accordance with this, CTA applicants also had, on average, 10.6 years of education - which is a full year lower than the averages of 11.6 years for CT applicants and 11.8 years for CJT applicants.

A comparison between CT and CJT course applicants reveals that CT courses attracted a higher proportion of male applicants, making up 63 % of the control group compared to 33 % for CJT. Moreover, 66 % of CT applicants were formally employed in 2012, with an average of 6 months of employment in the year, while only 48 % of CJT applicants were formally employed, with an average of 3 months of employment. Therefore, although many characteristics are similar between participants in CT and CJT courses, it appears that CT courses attracted a relatively higher number of individuals with recent formal work experience.

4.2. Balancing of applicants' characteristics: re-assigned treatment

In Section 4.1, we conclude that the original randomization was successfully implemented in the full sample, as well as within each course type. However, as previously discussed, if an applicant originally assigned to the treatment group was no longer available or dropped-out during the first 20 % of the training, the providers contacted applicants originally assigned to the control group and offered them a place in the program. In Table 3, we present balance tests for the sample after treatment reassignment. Since replacement was took place within training provider and course, we control for course fixed effects in all regressions.¹¹

¹¹ We control for course fixed effects, not course-by-gender fixed effects, as we do not know whether OTECs maintained gender stratification in the reassignment process.

Table 3
Experimental balancing, treatment after re-assignment.

	Full sample		CT		CJT		CTA	
	Control group (1)	Diff Treatment (2)	Control group (3)	Diff Treatment (4)	Control group (5)	Diff Treatment (6)	Control group (7)	Diff Treatment (8)
<i>Application characteristics</i>								
Male	0.349	−0.009 (0.012)	0.694	0.045 (0.032)	0.342	−0.019 (0.016)	0.201	−0.011 (0.017)
Age in years	33.111	−0.769** (0.354)	33.15	1.063 (0.941)	31.478	−1.067** (0.439)	36.75	−0.927 (0.755)
Poverty index (FPS)	5822.162	−84.537 (108.883)	6732.81	126.049 (342.108)	6064.903	−176.79 (140.016)	4843.73	39.629 (202.995)
Formal empl. 2011/12	0.604	−0.062*** (0.017)	0.735	−0.02 (0.044)	0.63	−0.083*** (0.022)	0.484	−0.033 (0.034)
Formal empl. 2012	0.481	−0.047*** (0.017)	0.66	0.006 (0.047)	0.509	−0.076*** (0.022)	0.334	−0.002 (0.032)
Months formal 2012	3.717	−0.336** (0.160)	6.095	0.537 (0.530)	3.775	−0.631*** (0.022)	2.451	−0.038 (0.285)
Labor income in 2012	133.000	−14.100** (5.988)	240.000	−6.374 (23.721)	132.000	−22.300*** (7.189)	84.611	2.225 (10.576)
<i>Household characteristics (baseline)</i>								
Married/partnership	0.452	−0.049*** (0.017)	0.592	0.022 (0.048)	0.446	−0.071*** (0.022)	0.396	−0.029 (0.033)
Household size	4.199	0.062 (0.060)	4.347	−0.065 (0.165)	4.177	0.007 (0.076)	4.179	0.260** (0.118)
Number of children	1.904	0.009 (0.039)	1.925	−0.058 (0.112)	1.861	0.024 (0.051)	1.99	0.003 (0.073)
Head of household	0.468	−0.028 (0.017)	0.592	0.033 (0.046)	0.401	−0.046** (0.022)	0.558	−0.013 (0.035)
HH members 0–5 yrs	0.429	−0.01 (0.017)	0.49	−0.049 (0.051)	0.404	−0.026 (0.022)	0.455	0.049 (0.035)
HH members > 65 yrs	0.141	0.004 (0.013)	0.122	−0.005 (0.034)	0.138	0.005 (0.016)	0.156	0.004 (0.026)
<i>Educational attainment and cognitive skills (baseline)</i>								
Years of education	11.457	0.02 (0.067)	11.626	0.014 (0.174)	11.806	−0.048 (0.075)	10.594	0.19 (0.17)
Literacy test	0.876	0.008 (0.007)	0.882	0.029 (0.020)	0.895	−0.007 (0.008)	0.832	0.033** (0.015)
Numeracy test	0.752	0.002 (0.012)	0.814	−0.052 (0.033)	0.763	0.017 (0.015)	0.696	−0.008 (0.026)
Spatial orientation	0.762	0.013 (0.015)	0.834	−0.014 (0.039)	0.765	0.034* (0.019)	0.721	−0.025 (0.032)
Fluid intelligence	54.132	0.295 (0.711)	53.842	1.886 (2.129)	56.209	−0.325 (0.877)	49.629	1.043 (1.463)
<i>Non-cognitive skills (baseline)</i>								
Locus of control	33.729	0.213 (0.165)	34.021	0.942* (0.548)	33.755	0.056 (0.201)	33.532	0.242 (0.331)
Rosenberg score	33.857	0.25 (0.183)	33.738	0.985* (0.572)	34.259	0.012 (0.221)	33.013	0.479 (0.382)
BIG 5: Extraversion	4.609	0.029 (0.052)	4.431	0.244* (0.141)	4.609	0.027 (0.066)	4.695	−0.071 (0.106)
BIG 5: Agreeableness	5.226	0.01 (0.047)	5.124	0.039 (0.137)	5.306	−0.059 (0.059)	5.094	0.164* (0.092)
BIG 5: Conscientious	5.885	−0.043 (0.046)	5.659	0.083 (0.125)	5.94	−0.084 (0.057)	5.87	−0.004 (0.095)
BIG 5: Neuroticism	5.033	0.102** (0.051)	5.048	0.088 (0.143)	5.135	0.028 (0.064)	4.799	0.292*** (0.104)
BIG 5: Openness	5.727	−0.016 (0.046)	5.71	0.036 (0.127)	5.755	−0.046 (0.058)	5.674	0.033 (0.095)
<i>Self-reported labor market status (baseline)</i>								
Paid employment	0.596	−0.030* (0.017)	0.836	0.028 (0.035)	0.517	−0.059*** (0.022)	0.658	0.013 (0.033)
Self-employment	0.229	0.019 (0.014)	0.185	−0.001 (0.038)	0.176	0.023 (0.017)	0.368	0.021 (0.033)
Labor income	100.227	−7.691** (3.888)	234.181	−4.774 (14.355)	78.545	−15.08*** (4.759)	83.396	9.506 (6.917)
Observations	4113		480		2732		901	
Joint significance (all)	p-value = 0.016		p-value = 0.634		p-value = 0.002		p-value=0.104	

Note: We control for course fixed effects in all regressions. Robust standard errors are reported in parentheses.

Analyzing the full sample after reassignment (columns 1 and 2), we now note significant differences in pre-treatment labor market outcomes. Specifically, those offered to participate in FOTRAB were less likely to have been formally employed or engaged in paid employment before the program. Additionally, they had fewer months of formal

employment in 2012 and lower labor income. This suggests that the reassignment process did not occur randomly, as initially intended. Instead, replacement individuals from the control group disproportionately consisted of disadvantaged individuals who were either unemployed or had lower earnings. Indeed, a joint significance test of

baseline characteristics in predicting treatment assignment is now statistically significant (p-value = 0.016).

However, apart from differences in pre-program labor market outcomes, the differences between those assigned to treatment and control groups are mostly small and statistically insignificant. This suggests that while baseline labor market outcomes may have been considered in the reassignment process, conditional on this, the reassignment was plausibly random. Consistent with this, we observe that when labor market outcomes are excluded, a joint test of all other baseline characteristics is not significantly different from zero (p-value = 0.253). This provides evidence that although selective reassignment occurred, it was driven by pre-treatment labor market outcomes, and, conditional on these characteristics, the reassignment process can be considered random.

When examining the treatment balance for each course type individually, we find evidence that while providers of CT and CTA courses may not have contacted SENCE/IDB, they likely adhered to the initial design by randomly selecting replacements from the list of controls. Indeed, for both individuals assigned to CT and CTA courses, a joint test of all baseline characteristics is not statistically significant.

In contrast, Table 3 suggests that providers of CJT courses deviated from the random reassignment process and instead chose disadvantaged control individuals. Indeed, analyzing column (6), we find that those assigned to participate had significantly worse pre-program labor market outcomes. However, as with the full sample, we find that, conditional on the pre-program labor market characteristics, the reassignment process was random. When labor market outcomes are excluded, a joint test of baseline characteristics on treatment status is not statistically significant (p-value=0.205).

In our empirical analysis, we incorporate the conditionally random reassignment process by controlling for labor market history in all regressions. Furthermore, we include controls for household characteristics and non-cognitive skills to address any other observed imbalances in Table 3.

4.3. Treatment take-up

Overall, among the 2968 individuals included in our analysis sample who were assigned to the FOTRAB program either initially or through reassignment, 70 % accepted the offer and successfully completed the first 20 % of the program, which we define as “participation” (further details in Section 5). Comparatively, 75 % (333) of individuals assigned to CT courses completed at least 20 % of the training, while CTA courses saw the highest participation rate, with 80 % of the 593 individuals completing this portion. In contrast, CJT courses had the lowest engagement, with only 66 % of the 2042 individuals completing the initial 20 % of the program.

In Appendix Table D.1, we present the characteristics of participants in the FOTRAB program compared to those who were assigned to a treatment but did not complete at least 20 % of the program. In the pooled sample, there are no differences in age, household size, number of children, educational attainment, and both cognitive and non-cognitive skills between participants and non-participants. However, participants were more likely to have recent formal employment, higher formal labor income, and more months of formal work experience. They were also more likely to be male, have a higher poverty score, and less likely to be married or cohabitating, and to live with young children. A joint test of baseline characteristics predicting participation is statistically significant (p-value = 0.000), suggesting that the decision to participate in the program was not random.

Further analysis by course type indicates that selection in participation is driven by individuals in the CJT courses. For CT and CTA courses, differences in baseline covariates are mostly insignificant, and a joint test of all characteristics is not statistically significant (p-values = 0.552 and 0.214, respectively). This suggests that, for these courses, the decision to participate in the program was plausibly random. However, in CJT courses, FOTRAB participants have worse pre-program formal

labor outcomes, are less likely to be married or head of the household, and are less likely to live with young children.

In general, the higher percentage of non-attendance in CJT courses might largely be attributed to their greater time commitment, which included 360 h of on-the-job training and job-finding support. However, the findings above suggest that participation in CJT courses might depend, in particular, on two factors. First, the fact that participants are negatively selected based on recent labor market history suggests that some may have dropped out upon receiving employment offers during training. This could have been especially relevant for individuals from poorer households, who might not be able to afford to overlook immediate job prospects. Second, the lower likelihood of participation among individuals who are married or have young children suggests that household responsibilities may hinder program completion. This is further supported by the observation that women are less likely to participate in FOTRAB, as previous research in Chile has shown that women disproportionately perform unpaid household tasks (OECD, 2021).

5. Empirical strategy

5.1. Parameters of interest and estimation

In this study, we estimate two parameters. First, the Intention-to-treat effect (ITT) identifies the effect of being randomly assigned to a training slot in FOTRAB. This is a highly relevant parameter from a policy perspective because, as for most active labor market policy programs, individuals can be offered to participate in FOTRAB but cannot be forced to participate. Thus, the ITT corresponds to the effect of making the program available to eligible individuals. In this study, when we refer to ITT estimates as the ‘impact of FOTRAB’, we are specifically addressing the impact of being assigned a place in the program. Especially in cases when it is expected that noncompliance will occur, the ITT shows whether a program is effective. We estimate the ITT using a specification of the form

$$Y_{ij} = \beta Z_i + X_i \lambda + \tau_j + \epsilon_{ij}, \quad (1)$$

where Y_{ij} is the outcome of applicant i to course j , Z_i is an indicator for being assigned to participate in the program either during original randomization or during the re-assignment process, X_i is a vector of individual controls, and τ_j are course-by-gender fixed effects. ϵ_{ij} is a random error term. Course-by-gender fixed effects account for the fact that randomization was implemented at course-level and by gender. We use the labor market information submitted at application, household characteristics, and non-cognitive skill variables as controls to account for selective reassignment, imbalances in these covariates, and to increase the precision of our estimates. The coefficient β gives a weighted average of gender-specific effects across the 123 different courses that were selected for the evaluation. The point estimate of the ITT and its standard error are presented in the first row in all panels of the result tables.

Second, we estimate the local average treatment effects (LATE) for compliers, i.e., for the subset of participants who comply with the assignment. Members of the control group had no access to participate in the program unless they were re-assigned to empty slots. Thus, we are facing a situation of one-sided non-compliance (e.g., Angrist and Pischke, 2009), which means that the effect for compliers is equal to the treatment effect on the treated (TOT). This is another policy-relevant parameter as it identifies the effect for those who participated in the program and for whom public resources were actually used. We estimate the LATE in the following specification by using the random assignment to the courses as an instrument for participation:

$$Y_{ij} = \beta D_i + X_i \lambda + \tau_j + \epsilon_{ij}, \quad (2)$$

where the participation dummy D_i is instrumented with the indicator for

assignment to the FOTRAB program Z_i in the first-stage equation

$$D_i = \pi Z_i + X_i \delta + \tau_j + \eta_{ij}. \quad (3)$$

Here, the participation dummy takes the value of one if a participant successfully completed at least the first 20 % of the program.

We estimate these equations by two-stage least squares using course-by-gender fixed effects in all equations. The coefficient β is a weighted average of course-by-gender specific LATE effects, defined as the impact of participation in FOTRAB on individuals who are induced to participate by being offered a training slot. This is equal to an up-scaling of the ITT by the first-stage effect of the instrument on the probability of participation (see Eq. (3)). In all the results tables, the coefficient and significance of the first-stage regressions are presented in the second row, while the LATE estimates are presented in the third row. We use robust standard errors in all specifications.¹²

5.2. Labor outcomes

Given the objectives of FOTRAB, the main outcomes of interest in the evaluation are the labor market outcomes of applicants. In both of the follow-up surveys, individuals were asked to report their employment status (i.e., paid employee or self-employed), their labor income, and whether social security contributions are paid.

Paid employment includes salaried workers in the public or private sector, and self-employment includes employers and workers on their own account. In Chile, paid workers and employers are required to contribute to the social security system. However, at the time of the evaluation, it was optional for own-account workers to contribute to the social security. Thus, we cannot interpret contribution to social security as a direct measure of job formality.

We complement the analysis with employment records from administrative data. We use data from the unemployment insurance (UI) system, which contains employment and earning histories of formal workers in the country.¹³ This data is administered by the Unemployment Fund Administrator and contains records from all formal dependent workers until four years after application.¹⁴

6. The impact of FOTRAB on labor outcomes

In Table 4, we present the program's impact on the probability of paid employment, self-employment, and labor income in both the first and second follow-ups. Being assigned to and participating in FOTRAB has no effect on the likelihood of paid employment or self-employment in either follow-up. We find that the program leads to a significant increase in labor income in the first follow-up survey. However, this effect diminishes over time and is no longer present in the second follow-up survey. The increase in monthly income is 12,200 Chilean Pesos for those assigned to treatment and 16,420 Pesos for program participants, roughly equivalent to 25 and 33 US\$, respectively.

Given the differential and selective attrition observed in the first follow-up survey, in Appendix Table F.1, we examine the potential effects that attrition may have had on our ITT estimates on labor income in the first follow-up survey. In column (1), we estimate and implement attrition propensity weights (e.g., Baird et al., 2019), first running a probit regression predicting presence in the first follow up survey with baseline covariates and their interaction with treatment. Relevant

baseline covariates are chosen using a Lasso estimator. We then weight outcomes by the inverse of this follow-up success probability. In columns (2) and (3), we follow Kling and Liebman (2004) and impute to the missing observations the mean within their respective treatment group ± 0.05 *standard deviation. We find that IPW-adjusted estimates are very close to our original estimates, and the Kling and Liebman bounds present a very consistent picture of a short-term, positive impact on labor income.

To complement the survey findings, we show the program's effect using administrative records in Table 5. In the analysis, we use annual employment and income measures to average out the noisiness in the monthly data. Consequently, the estimated treatment effects reflect the impact of assignment to FOTRAB on the average proportion of months spent in formal employment per year, as well as the average annual formal income earned.

The results indicate a significant negative lock-in effects during the first year on formal employment and income for participants assigned to the program, likely due to lower search intensity during training. After the first year after applying to FOTRAB, the effects increase but never turn positive over the entire observational period. Comparing the results in Tables 4 and 5, we find that the positive short-term effects on labor income observed in the survey data are not present in the administrative data.

The self-reported data collected in the survey on paying social security contributions is most comparable to formal employment in the administrative data, which refers to those contributing to UI. Therefore, in Table H.1 in the Appendix, we present results on the probability of paying social security contributions and other reported survey outcomes such as hours worked, probability of unemployment, and probability of inactivity (i.e., not participating in the labor market). We find that, similar to Table 5, the point estimates of the program impact on the probability of paying social security are negative and not statistically significant. Since both the administrative data and contributions to social security capture formal employment, together, these suggest that the additional income reported in Table 4 may be due to increased earnings from informal activities. However, both data sources consistently show that, in the long run, there are no significant effects on employment or income.¹⁵

6.1. Impact of FOTRAB by course type

We next examine whether the effect of FOTRAB varies across the three different course types outlined in Section 2. Given that we randomized individuals to treatment or control within each course type separately, we have a valid control group for each course type. Panel A of Table 6 presents the impacts of the most basic type (CT courses), involving only classroom training. Although we observe positive ITT and TOT estimates on the probability of paid employment in both follow-up surveys, these coefficients are not statistically significant. However, there is a statistically significant positive effect on long-term labor income, with monthly increases of 35,570 Chilean Pesos for applicants and 43,880 Pesos for participants (equivalent to 72 and 149 US\$, respectively).¹⁶

In Panel A of Table 7, we report the results for CT courses obtained using administrative data. Consistent with Table 6, we find no significant effect on employment throughout the period. However, despite

¹² Appendix E details the assumptions required for our LATE analysis.

¹³ UI is an individual savings account for each formal worker. Both the worker and the employer contribute to this fund, although UI is supplemented by the Solidarity Fund, which is financed by public and private (employer) contributions.

¹⁴ If an applicant does not appear in the dataset in a given month, it means that she/he was not formally employed in that month. Consequently, for these periods, we assign the participant a formal income of zero.

¹⁵ As a robustness check, we conduct a pre-treatment outcome evaluation for the four years preceding training, allowing us to identify any effects prior to training (Ashenfelter and Card, 1985), or other selectivity based on pre-treatment labor outcomes. The results show that the effects are zero over all four pre-training years. (Table G.1 in the Appendix).

¹⁶ The findings on other outcomes reported in Table H.2 show that applying and participating in CT courses significantly reduces the probability of unemployment.

Table 4
Effects on labor market outcomes.

	Paid-employment		Self-employment		Labor income	
	FU 1	FU 2	FU 1	FU 2	FU 1	FU 2
(1) ITT	0.010 (0.017)	−0.012 (0.016)	0.001 (0.014)	0.010 (0.014)	12.20** (5.47)	−0.79 (5.53)
(2) First stage	0.744*** (0.010)	0.747*** (0.009)	0.744*** (0.010)	0.747*** (0.009)	0.743*** (0.010)	0.746*** (0.010)
(3) LATE=TOT	0.014 (0.023)	−0.016 (0.024)	0.002 (0.018)	0.013 (0.018)	16.42** (7.16)	−1.06 (7.86)
(4) Control means	0.528	0.548	0.196	0.185	172.12	207.21
Num. of obs.	4101	4055	4095	4053	3951	3906

Note: This table reports the impact of being assigned to and participating in FOTRAB on different forms of employment and labor income measured in 1000 CLP. FU 1 and FU 2 are the abbreviations for the first and second follow-ups, respectively. The first and second follow-up surveys were collected 18 and 35 months after individuals applied to FOTRAB, respectively. The TOT estimates are obtained by using the random assignment as an instrument for participation. Robust standard errors are reported in parentheses. All regressions control for course-by-gender fixed effects, basic characteristics provided at application, household characteristics, and non-cognitive skills measured at the baseline. All missing covariates are set to zero and dummies for missing values are included. Significance levels are indicated as: *** 1 %, ** 5 %, and * 10 %.

Table 5
Impact of assignment to FOTRAB (using administrative records).

	Year 1	Year 2	Year 3	Year 4
<i>Formal employment</i>				
(1) ITT	−0.078*** (0.012)	−0.005 (0.013)	−0.004 (0.013)	−0.020 (0.014)
(2) Control means	0.401	0.433	0.440	0.457
<i>Formal income</i>				
(1) ITT	−27.11*** (4.92)	−6.75 (5.95)	−2.44 (6.96)	−9.08 (7.71)
(2) Control means	134.07	169.17	187.75	211.14
Num. of obs.	4608	4608	4608	4608

Note: This table reports the impact of being assigned to FOTRAB on yearly average formal employment and labor income measured using administrative data. Year 1 refer to the first year after participant applied to the program, and all subsequent years follow an analogous meaning. Robust standard errors are reported in parentheses. All regressions control for course-by-gender fixed effects, basic characteristics provided at application, household characteristics, and non-cognitive skills measured at the baseline. Significance levels are indicated as: *** 1 %, ** 5 %, and * 10 %.

positive point estimates, we find no significant effect on formal labor income in Table 7. This suggests that at least part of the additional long-term income observed for those assigned to CT courses might have come from informal activities. Unlike before, we find no lock-in period for CT courses, likely due to their design for individuals seeking to combine work and training.

We next investigate the effects of being assigned to and participating in CJT courses, which include both classroom training and job training (Panel B of Tables 6 and 7). Using survey data (Table 6), we find a significant, negative ITT and TOT effect on paid employment in the long-term. Participation in CJT courses reduced the probability of being a paid worker by 7 pp, with no discernible effect on self-employment. This negative effect is also reflected in significantly reduced labor income in the second follow-up for those assigned to training, with a decrease of 11,560 Chilean Pesos (equivalent to 23 US\$). The same trend is observed using administrative data, as depicted in Table 7. Following a negative lock-in period, where the proportion of months in formal employment decreases by about 10 pp and average monthly formal income declines by nearly 32,000 Pesos (64 US\$), the effects increase to close to zero. However, in the longer term (approximately four years post-application), the effects on formal labor market outcomes become significantly negative, aligning with the survey findings.

Finally, we assess the impact of CTA courses, which involve classroom training complemented with technical assistance, targeting applicants seeking to be self-employed. The program effects for these courses are detailed in Panel C of Tables 6 and 7. While we observe

positive point estimates on the probability of self-employment, they are not statistically significant. Overall, the results suggest participants may have attempted to substitute paid employment with self-employment, at least in the short term. Analyzing administrative records (Table 7), we observe significant lock-in effects in the first year after application. However, consistent with the survey findings, we observe no significant effects on employment or income in the long-term.

6.2. Cost-benefit analysis

Examining the impacts of the three course types, we observe that only the CT program has any significant positive effect on labor income. Consequently, we present a simple calculation of a cost-benefit analysis for CT courses. For this, we consider the effects on earnings and use a discount rate of 5 percent per year to estimate the Net Present Value. According to Table 6, we estimate zero benefit of CT courses in the initial follow-up, and US\$72 per month (35,570 Chilean Pesos) in the subsequent follow-up, which implies an annual benefit of US\$864.

Following Attanasio et al. (2015) and Novella and Valencia (2022), we estimate the impact on earnings for 5 years after the application to the training program. Considering the timing of the follow-up surveys (18 and 36 months after application, respectively), we assume that the null impact of CT courses applies for the first two years, and the positive benefit persists for the following three. The calculation considers two scenarios: one, where gains are permanent; and the other, where gains depreciate at an annual rate of 10 percent. We also assume that the program does not affect the growth rate of earnings.

Regarding costs associated with the program, the average cost per beneficiary and hour of training was US\$2.40. Given an average course duration of 500 h, this implies a cost of US\$1200 per beneficiary. Additionally, each participant received a daily subsidy of US\$6, with an average course duration of 125 days, totaling US\$750 in subsidies. Consequently, the total direct cost per beneficiary for CT courses was US \$1950. Additionally, an estimate from the IDB approximates the indirect cost at \$813 per beneficiary. This cost includes the opportunity cost incurred by trainees for being engaged in training and not in employment; it is computed by multiplying the average hourly earnings of the four poorest income deciles by the number of hours invested in the training course.

In the first scenario, the internal rate of return (IRR) is calculated at −5 %, while it is −8 % for the second scenario (with depreciation). This negative IRR is substantially lower than IRRs found for other job training programs in LAC countries. For example, Attanasio et al. (2011) reported an IRR ranging from 21 to 35 % for *Jóvenes en Acción*, an ALMP for vulnerable youth in Colombia, and Novella and Valencia (2022) found an IRR between 9 and 12 % for *Programa de Apoyo al Empleo*, a public program in Bolivia offering job seekers information about

Table 6
Effects on labor outcomes by course type.

	Paid-employment		Self-employment		Labor income	
	FU 1	FU 2	FU 1	FU 2	FU 1	FU 2
Panel A: CT courses						
(1) ITT	0.040 (0.070)	0.075 (0.052)	−0.002 (0.044)	−0.024 (0.048)	19.13 (15.60)	35.57*** (12.46)
(2) First stage	0.795*** (0.026)	0.811*** (0.026)	0.795*** (0.026)	0.812*** (0.026)	0.801*** (0.027)	0.811*** (0.027)
(3) LATE=TOT	0.050 (0.057)	0.092 (0.058)	−0.002 (0.044)	−0.029 (0.048)	24.17 (20.53)	43.88** (20.44)
(4) Control means	0.667	0.639	0.177	0.203	275.42	299.70
Num. of obs.	478	484	478	484	461	461
Panel B: CJT courses						
(1) ITT	0.021 (0.021)	−0.046** (0.021)	−0.016 (0.016)	0.014 (0.015)	10.16 (7.00)	−11.56* (7.02)
(2) First stage	0.699*** (0.012)	0.698*** (0.012)	0.698*** (0.012)	0.698*** (0.012)	0.698*** (0.012)	0.699*** (0.012)
(3) LATE=TOT	0.030 (0.032)	−0.067** (0.033)	−0.023 (0.023)	0.020 (0.023)	14.57 (9.53)	−16.56 (10.93)
(4) Control means	0.539	0.579	0.158	0.139	163.60	206.05
Num. of obs.	2727	2680	2722	2678	2630	2593
Panel C: CTA courses						
(1) ITT	−0.024 (0.037)	0.036 (0.029)	0.040 (0.033)	0.012 (0.035)	16.72 (12.19)	12.04 (11.92)
(2) First stage	0.814*** (0.017)	0.817*** (0.017)	0.814*** (0.017)	0.817*** (0.017)	0.811*** (0.018)	0.813*** (0.018)
(3) LATE=TOT	−0.029 (0.040)	0.044 (0.040)	0.049 (0.039)	0.015 (0.036)	20.39 (12.79)	14.62 (12.55)
(4) Control means	0.438	0.433	0.290	0.279	141.56	163.67
Num. of obs.	896	891	895	891	860	852

Note: This table reports the impact of being assigned to and participating in one of the course types on different forms of employment and labor income measured in 1000 Chilean Pesos. FU 1 and FU 2 are the abbreviations for the first and second follow-ups, respectively. The first and second follow-up surveys were collected 18 and 35 months after individuals applied to FOTRAB, respectively. The TOT estimates are obtained by using the random assignment as an instrument for participation. Robust standard errors are reported in parentheses. All regressions control for course-by-gender fixed effects, basic characteristics provided at application, household characteristics, and non-cognitive skills measured at the baseline. All missing covariates are set to zero and dummies for missing values are included. Significance levels are indicated as: *** 1 %, ** 5 %, and * 10 %.

vacancies posted by formal firms.

We do not present a formal cost-benefit analysis for CTA and CJT courses, as the estimated effects of these courses on earnings are zero and negative, respectively. Given that CTA courses incurred an estimated direct cost of \$2450 per beneficiary, and CJT courses had an estimated direct cost of \$2220 per beneficiary, it becomes apparent that these courses were highly cost ineffective.¹⁷ Taken together, these findings suggest that even for CT courses, which exhibit the largest positive effect on labor income, the costs associated with the courses offered under FOTRAB likely exceed any positive benefits yielded by the program.

7. Heterogeneous impact of FOTRAB on labor outcomes

In Section 6, we find that, on average, the FOTRAB program shows limited effectiveness on labor market outcomes. However, it is possible that these average effects belie meaningful heterogeneity. Consequently, in this section, we analyse heterogeneous effects of the program to understand whether there are certain subgroups for which the program made a larger difference.

¹⁷ The cost per beneficiary for CTA courses comprises US\$1,950 for classroom training, an additional \$96 for 40 hours of technical assistance, and a supplementary US\$404 subsidy designated for purchasing tools. The cost per beneficiary for CJT courses includes US\$1,950 for classroom training, alongside a daily subsidy of \$6 for 45 days of on-the-job training at a firm (assuming an 8-hour workday).

7.1. Subgroup analysis by gender, age group, and employment prospects

First, we analyze whether the program's effectiveness varies by gender, age-groups, and employment prospects in Table 8. In Panel A, we estimate the effect of being offered training for the subgroup of women, and the interaction indicates if the training is more or less effective for males compared to women. We find that men who were offered training earn significantly more than women in both the short- and long-term. The difference between male and female long-term paid employment is also large and positive, but does not reach statistical significance. These findings contrast with the majority of results on labor program effectiveness in LAC (Escudero et al., 2019) and worldwide (Card et al., 2018), that find that many programs are more effective for women. However, they are in line with the findings of Alzúa et al. (2016) on a labor market program for youth in Argentina.

In Panel B, we investigate heterogeneity by age-group. We find that the program is less effective for youth (i.e., younger than 30) than for adults. The interaction term coefficients have negative signs for almost all outcomes, and reach statistical significance for short-term paid employment and labor income. This finding is in line with the evidence of training programs for OECD countries (Kluve et al., 2019). However, it contrasts with the evidence for LAC and with the findings from non-experimental evaluations of the programs that preceded FOTRAB (Jara, 2001; Centro Microdatos, 2006, Centro Microdatos, 2008; Ministerio de Trabajo y Previsión Social, 2011).

Lastly, in Panel C, we investigate heterogeneity according to baseline formal labor prospects. To assess this, we run a logistic regression of whether the participant was formally employed sometime in 2011/12 on a vector of individual baseline characteristics. From this, we calculate the predicted probability of recent formal work experience for each

Table 7

Effects on formal labor outcomes by course type (using administrative records).

	Year 1	Year 2	Year 3	Year 4
Panel A: CT courses				
<i>Formal employment</i>				
(1) ITT	0.036 (0.030)	0.046 (0.044)	0.042 (0.043)	0.046 (0.051)
(2) Control means	0.576	0.567	0.555	0.539
<i>Formal income</i>				
(1) ITT	2.13 (13.87)	4.85 (21.16)	11.88 (24.02)	8.62 (32.58)
(2) Control means	240.25	272.00	294.04	319.11
Num. of obs.	569	569	569	569
Panel B: CJT courses				
<i>Formal employment</i>				
(1) ITT	−0.096*** (0.015)	−0.006 (0.017)	−0.013 (0.016)	−0.049*** (0.018)
(2) Control means	0.422	0.464	0.468	0.500
<i>Formal income</i>				
(1) ITT	−31.55*** (5.97)	−4.29 (7.58)	−5.20 (8.87)	−18.73* (9.83)
(2) Control means	132.91	175.36	193.72	193.72
Num. of obs.	2892	2892	2892	2892
Panel C: CTA courses				
<i>Formal employment</i>				
(1) ITT	−0.059** (0.023)	−0.006 (0.026)	0.015 (0.023)	0.019 (0.022)
(2) Control means	0.262	0.291	0.315	0.315
<i>Formal income</i>				
(1) ITT	−16.86** (7.32)	−2.68 (9.32)	6.12 (10.75)	4.88 (10.38)
(2) Control means	80.07	100.29	117.52	135.64
Num. of obs.	934	934	934	934

Note: This table reports the impact of being assigned to one of the course types on yearly average formal employment and labor income measured using administrative data. Year 1 refer to the first year after participant applied to the program, and all subsequent years follow an analogous meaning. Robust standard errors are reported in parentheses. All regressions control for course-by-gender fixed effects, basic characteristics provided at application, household characteristics, and non-cognitive skills measured at the baseline. Significance levels are indicated as: *** 1 %, ** 5 %, and * 10 %.

participant, based on their baseline characteristics. Subsequently, we divide individuals into two groups according to this probability. Those whose estimated probability falls below the average are categorized as individuals with lower ex-ante formal labor prospects. Analyzing Panel C, we find that individuals with lower prospects earn significantly more than those with high prospects, in the short-term. This is consistent with the evidence in [Card et al. \(2018\)](#) that finds that many training programs tend to have larger effects for the long-term unemployed. However, this differential does not persist in the long-term (although the interaction coefficient remains negative).

7.2. The quality of training providers and courses

We next investigate whether the program's effectiveness correlates with training provider and course quality. The training providers of FOTRAB - called OTECs - are non-profitable organizations devoted to the training of vulnerable people. To provide publicly-sponsored courses, they have to meet requirements with respect to infrastructure and staff. However, we observe substantial variation in the implementation of these requirements, which allows us to construct a provider quality index.

In order to create such an index, we rely on course level information submitted by the providers to SENCE when applying for funding, and detailed provider characteristics obtained from survey data at the OTEC-

Table 8

Effects on labor outcomes by gender and age-group.

	Paid-employment		Self-employment		Labor income	
	FU 1	FU 2	FU 1	FU 2	FU 1	FU 2
Panel A: Effects by gender						
ITT female	0.020 (0.021)	−0.028 (0.019)	−0.006 (0.018)	0.013 (0.017)	4.49 (5.51)	−7.96 (6.35)
ITT × male	−0.026 (0.038)	0.053 (0.035)	0.017 (0.028)	−0.012 (0.030)	20.48* (12.22)	24.24** (10.97)
Panel B: Effects by age-group						
ITT adult	0.034 (0.023)	−0.013 (0.023)	0.006 (0.019)	0.023 (0.018)	21.94*** (7.62)	3.11 (7.64)
ITT × young	−0.056* (0.031)	0.002 (0.037)	−0.010 (0.026)	−0.029 (0.025)	−22.56** (10.98)	−8.43 (11.81)
Panel C: Effects by baseline formal labor prospects						
ITT lower prospects	0.033 (0.024)	−0.011 (0.022)	0.013 (0.019)	0.013 (0.016)	20.90*** (7.38)	6.59 (7.89)
ITT × higher prospects	−0.047 (0.034)	0.003 (0.034)	−0.027 (0.026)	−0.011 (0.028)	−20.70* (10.95)	−14.76 (9.72)

Note: This table reports the heterogeneous effects of being assigned to FOTRAB. Labor income is measured in 1000 CLP. FU 1 and FU 2 are the abbreviations for the first and second follow-ups, respectively. The first and second follow-up surveys were collected 18 and 35 months after individuals applied to FOTRAB, respectively. Robust standard errors are reported in parentheses. We control for course fixed effects, basic characteristics provided at application, household characteristics, and non-cognitive skills measured at the baseline. All missing covariates are set to zero and dummies for missing values are included. Significance levels are indicated as: *** 1 %, ** 5 %, and * 10 %.

level. In 2013, an OTEC questionnaire was administered to the 12 OTECs where the 123 courses that are part of the evaluation took place. We provide summary statistics of the variables used to construct a quality index in [Table 10](#). On average, classes consist of 21.4 persons, teachers have 2.2 years of teaching experience and 2.6 years of working experience, and teaching hours amount to 387 for technical skills and 95 for transversal abilities. More than half of the OTECs report that they invest in teacher training. The average share of teachers with a university degree is 44 %, and, on average, teachers have teaching experience of 5 years. About 20 % of providers use psychological tests in the recruitment interviews and almost half of them test students before the course starts.

We combine these components to build a quality index, constructed via principal component analysis. We categorize the 123 courses into two groups based on this index: those below the average within their course type are deemed low-quality, while those above are considered high-quality. We show course level and provider characteristics according to quality level in [Table 9](#) and find significant differences between low- and high-quality courses and providers. For example, high-quality courses have a lower capacity, teachers have more experience, they invest more frequently in teacher training, and more often use testing procedures. Overall, the values for low- and high-quality courses are very plausible and support the view that the constructed index successfully separates low- and high-quality courses from each other.

Finally, we investigate heterogeneous program effects by quality in [Table 10](#). Since the applicants were not randomly allocated across courses of different quality, the reported results reflect mere correlations. We find significantly larger effects for applicants assigned to a high-quality course on labor income in the longer run. These results are consistent with the findings of [Galdo and Chong \(2012\)](#), which is the only other paper analyzing the differential effect of training programs by provider quality.

8. What might explain the null effects?

Although we find some positive effects of FOTRAB on labor income

Table 9

Variables used to construct quality measure.

	All	Low	High	Diff
Panel A: Course-level information				
Course capacity	21.42 (0.45)	22.00 (1.20)	20.97 (3.11)	−1.03** (0.45)
Years of teaching experience of teachers	2.16 (0.07)	2.00 (0.00)	2.29 (0.55)	0.29*** (0.07)
Years of working experience of teachers	2.63 (0.11)	2.09 (0.29)	3.06 (0.75)	0.97*** (0.11)
Number of teaching hours in technical abilities	387.12 (25.51)	411.74 (202.91)	367.86 (54.66)	−43.89* (25.51)
Number of teaching hours in transversal abilities	95.06 (6.96)	80.07 (28.85)	106.78 (44.27)	26.71*** (6.96)
Panel B: Provider-level information				
Investments in teacher training	0.54 (0.07)	0.15 (0.36)	0.84 (0.37)	0.69*** (0.07)
Share of teachers with university degree	0.44 (0.07)	0.06 (0.23)	0.74 (0.44)	0.68*** (0.07)
Years of experience of teachers	4.98 (0.37)	4.22 (2.38)	5.57 (1.70)	1.34*** (0.37)
Use of psychological tests in teacher recruitment	0.20 (0.07)	0.06 (0.23)	0.32 (0.47)	0.26*** (0.07)
Test students' skills before course start	0.46 (0.07)	0.09 (0.29)	0.74 (0.44)	0.65*** (0.07)

Note: The standard deviations of the means are reported in parentheses. The difference between the characteristics of low- and high-quality training courses are calculated from a regression of the characteristics on the indicator for high quality. The standard error of the difference is reported in parentheses. Significance levels are indicated as: *** 1 %, ** 5 %, and * 10 %.

Table 10

Effects on labor outcomes by provider quality.

	Paid employment		Self-employment		Labor income	
	FU 1	FU 2	FU 1	FU 2	FU 1	FU 2
ITT lower quality	0.015 (0.026)	−0.020 (0.024)	−0.007 (0.020)	0.006 (0.019)	6.081 (8.084)	−11.83 (8.535)
ITT × higher quality	−0.026 (0.033)	0.012 (0.033)	0.032 (0.027)	0.014 (0.026)	3.202 (10.609)	19.70* (11.824)

Note: This table reports the effects of being assigned to FOTRAB for applicants in lower quality OTECs and the difference between to higher quality OTECs on different forms of employment and labor income measured in 1000 Chilean Pesos. FU 1 and FU 2 are the abbreviations for the first and second follow-ups, respectively. The first and second follow-up surveys were collected 18 and 35 months after individuals applied to FOTRAB, respectively. Robust standard errors are reported in parentheses. We control for course fixed effects, basic characteristics provided at application, household characteristics, and non-cognitive skills measured at the baseline. All missing covariates are set to zero and dummies for missing values are included. Significance levels are indicated as: *** 1 %, ** 5 %, and * 10 %.

for CT courses and certain subgroups, the effects on most labor market outcomes are essentially zero. This stands in contrast to international evidence which find that training programs tend to have positive average effects on a number of labor market outcomes. Below, we discuss two mechanisms that might be behind the disconnect between our results and the broader literature.

8.1. Limited impact of FOTRAB on skill outcomes

Existing literature, both internationally (Card et al., 2018) and in LAC (Escudero et al., 2019), has emphasized that one of the primary

mechanisms through which training programs are expected to improve labor market outcomes is through increasing human capital accumulation. Therefore, in this section, we assess whether the FOTRAB program successfully improved the skill development of applicants and participants.

As discussed in Section 3.2, several tests for cognitive and non-cognitive skills were collected in the follow-up surveys. For the first follow-up, we organize the test scores into summary indices of cognitive and non-cognitive skills. First, we standardize each individual skill outcome by the control group mean and the standard deviation. Then, we summarize the standardized test scores to one cognitive skill index and one non-cognitive skill index in the first follow up (similar indices are used in the early childhood education literature, e.g., Walters, 2015; Deming, 2009). The test scores in the second follow-up are standardized in the same way, however, since only one cognitive and non-cognitive test was implemented, the skill outcomes in the second and first follow-up surveys cannot be directly compared over time.

In Table 11, we report the effects of FOTRAB on skills. We find that while the program has a marginally significant effect on the cognitive skill index in the first follow-up, it has no persistent effects on the CRT (cognitive reflection test), measured three years after application, nor any of the non-cognitive skill measures in the either of the follow-up surveys. Therefore, although FOTRAB aimed at improving workers' skills, the program seems to have been largely ineffective in developing most measured cognitive and non-cognitive skills. The low effectiveness of FOTRAB on skill improvement offers one explanation for the overall modest impact on labor market outcomes.

Several factors could explain the limited impact of FOTRAB on human capital accumulation, including the relevance of the training content (discussed below). However, another possibility is that the program employed relatively inexperienced teachers. Table 10 indicates that the average years of teaching experience among instructors in FOTRAB courses were significantly lower than to the average across their providers. This suggests that the program may have relied more heavily on relatively inexperienced educators, which has been found to be negatively associated with student achievement gains in other

Table 11

Effects on skill outcomes.

	Cognitive skills		Non-cognitive skills	
	Index FU 1	CRT FU 2	Index FU 1	Self-efficacy FU 2
(1) ITT	0.036* (0.021)	0.010 (0.035)	−0.035 (0.026)	0.007 (0.030)
(2) First stage	0.743*** (0.009)	0.746*** (0.009)	0.743*** (0.009)	0.743*** (0.009)
(3) LATE=TOT	0.048* (0.030)	0.014 (0.047)	−0.048 (0.036)	0.010 (0.037)
(4) Control means	−0.012	−0.039	0.026	0.007
Num. of obs.	4107	4045	4107	4590

Note: This table reports the impact of being assigned to and participating in FOTRAB on different skill outcomes. The first follow-up indices correspond to the average of the z-scores (the score on each skill test was standardized taking into account the control group mean and the standard deviation) of the cognitive (literacy, numeracy, spatial orientation, and fluid intelligence) and non-cognitive (self-esteem, CPS, and grit) tests included. The test scores (CRT and self-efficacy) in the second follow-up are standardized in the same way. FU 1 and FU 2 are the abbreviations for the first and second follow-ups, respectively. The first and second follow-up surveys were collected 18 and 35 months after individuals applied to FOTRAB, respectively. CRT stands for cognitive reflection test. The TOT estimates are obtained by using the random assignment as instrument for participation. Robust standard errors are reported in parentheses. All regressions control for course-by-gender fixed effects, basic characteristics provided at application, household characteristics, and non-cognitive skills measured at the baseline. All missing covariates are set to zero and dummies for missing values are included. Significance levels are indicated as: *** 1 %, ** 5 %, and * 10 %.

settings (Kini and Podolsky, 2016).

8.2. Supply-driven program design

One of the primary objectives of job training programs is to impart participants with practical, job-relevant skills that can readily be applied in the workplace. As such, ensuring the relevance of these skills to the local labor market is crucial for their effectiveness. An increasing number of training programs are adopting a market-based approach in their design, often referred to as a “demand-driven” model (Alzúa et al., 2020; Almeida, Behrman, and Robalino, 2012). Such programs have become widely used in several Latin American countries, and existing evidence of their effectiveness shows positive effects on employment and earnings (e.g., Alzúa et al., 2020; Attanasio, Kugler, and Meghir 2011; Betcherman et al., 2007).

In contrast, the FOTRAB program followed a “supply-driven” approach. In this model, after SENCE selected participating OTECs, providers had complete autonomy over the content taught to participants. There was no consultation with industry partners to ensure that skills taught in the program were relevant to the evolving demands of the local workforce. Instead, providers simply offered training based on their existing expertise.

Without engaging with local industry partners to understand their specific skill requirements, the FOTRAB program may have failed to equip participants with the skills most valued and demanded in the labor market. As a result, even if individuals acquired technical skills through the program, there was no guarantee of demand for these skills in the job market, either immediately after program completion or in the long term. This disconnect offers another plausible explanation for the overall modest impact of the FOTRAB program on labor market outcomes.

9. Discussion and conclusion

The experimental implementation of one cohort of the FOTRAB program offers the first opportunity to evaluate the causal effects of a large publicly-funded job training program in Chile. The experimental design, together with access to survey and administrative data, allowed us to measure the effects of FOTRAB on labor market outcomes four years after application.

Our findings indicate that being assigned to and participating in FOTRAB has no effect on the probability of engaging in paid employment or being self-employed. While we observe a short-term increase in labor income, this improvement disappears over time. Analyzing the effect according to course type, we find that classroom training improves long-term labor income and reduces the probability of being unemployed. Courses combining classroom training and technical assistance suggest some improvements in the probability of being self-employed in the short run, although the estimates are not statistically significant. However, courses combining classroom and on-the-job training show no effectiveness. Further analysis reveals that the program’s positive effects on labor income are strongest for males, older individuals, and those with lower estimated formal labor prospects. Additionally, training from high-quality providers has a larger positive effect on income.

Taken together, our results suggest that, overall, the FOTRAB program has had limited effectiveness on labor market outcomes, potentially due to several factors. First, although the program devoted a considerable proportion of the training to skills development, we find almost no evidence that FOTRAB improved the cognitive and non-cognitive skills measured in the evaluation. However, it is important to note that our evaluation only measured a limited range of skills. This limitation means that the FOTRAB program may have positively impacted other skills not captured in our evaluation. Nevertheless, if such improvements did occur, they do not appear to have translated into better labor market outcomes. Second, unlike many successful training programs in the region, the FOTRAB program operated on a “supply-

driven” model, whereby providers had full autonomy over the curriculum without consultation with industry partners to ensure alignment with the changing needs of the local job market.

One limitation of our analysis is related to the fact that, given their high vulnerability, individuals in the FOTRAB sample might have been enrolled in other programs during the analysis period. Given the data collected, we are unable to definitively rule out that the observed effects are not affected by substitution bias. As a robustness check, however, we re-estimated the main results controlling for whether, in the first follow up survey, respondents reported participating in any other job training programs since 2013. The results are robust to this alternative specification.¹⁸

While the FOTRAB program has been discontinued, SENCE continues to provide training programs in Chile with similar designs. For example, following a shift in government administration in 2014, the government prioritized the implementation of a new training program – *Más Capaz*. The design of *Más Capaz* was very similar to FOTRAB, prioritizing the training of vulnerable populations. It is unlikely that *Más Capaz*, or any other analogous programs, have been developed in response to the perceived (in)effectiveness of FOTRAB, as there were no comprehensive evaluations of the program when *Más Capaz* was initiated. Consequently, we believe that our findings hold a number of valuable insights for informing the future design of training programs in Chile.

First, future programs could benefit from closer alignment with labor market demand to shape program curricula effectively (Alzúa et al., 2020). Collaboration with industry partners in the design phase could help identify sectors with unmet workforce needs, ensuring that participants acquire technical skills that are highly sought after by employers. Second, establishing and supervising better quality standards for training providers and educators is crucial (Galdo and Chong, 2012). For FOTRAB, the courses were taught by relatively inexperienced instructors, and we found no evidence that the non-cognitive skills component was taught by accredited professionals.

Third, the lower participation of women and individuals with young children suggests that complementary interventions, such as childcare provision, may enhance training completion rates. Fourth, as with other SENCE programs (Novella et al., 2017), future programs might benefit from offering vocational orientation and labor market information (e.g., returns to different types of training) aimed at improving the match between applicants and the supply of training courses. Lastly, expanding the target population to include individuals from diverse backgrounds may improve program effectiveness, as evidenced by previous non-experimental evaluations of similar programs (Duflo et al., 2011). While our findings hold particular importance for shaping future training programs in Chile, certain insights may also prove relevant to programs across LAC and more broadly.

Beyond these design considerations, we believe that our evaluation holds important lessons for improving experiments aimed at evaluating the impacts of job training programs. In particular, the unanticipated deviation among training providers in the reassignment process underscores the importance of establishing and conducting thorough quality checks and controls throughout every stage of the experiment. Additionally, our results highlight that it is important measure the impacts over longer-time frames. Most studies measure impacts over at most 1–2 years (McKenzie, 2017), leaving the authors to speculate about cost-benefit on the basis of assumptions about how impacts vary over time. Our findings show that tracking impacts over longer time periods is needed, as the estimated effects may differ substantially over time, and that linking participants to administrative records offers one promising way to do this.

¹⁸ Full results are reported in Tables I.1 and I.2 in the Appendix.

CRedit authorship contribution statement

Annabelle Doerr: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation. **Rafael Novella:** Writing – review & editing, Writing – original draft, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Data availability

The authors do not have permission to share data.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.labeco.2024.102619](https://doi.org/10.1016/j.labeco.2024.102619).

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